Human Reproduction

Humans are sexually reproducing and viviparous, and the reproductive events in humans run from the forming of gametes to the delivery of the baby.

The sequence of events

- formation of gametes, that is gametogenesis, sperms in males and ovum in females
- transfer of sperms into the female genital tract, that is insemination, and fusion of male and female gametes, that is fertilisation, leading to formation of a zygote
- formation and development of the blastocyst and its attachment to the uterine wall, that is implantation
- embryonic development, that is gestation, and delivery of the baby, that is parturition

Where male and female differ

- these reproductive events occur after puberty
- there are remarkable differences between the reproductive events in the male and in the female
- sperm formation continues even in old men, but formation of ovum ceases in women around the age of fifty years

The male reproductive system

- it is located in the pelvis region
- it includes a pair of testes along with accessory ducts, glands and the external genitalia

In this chapter

2.1 The male reproductive system
2.5 The female reproductive system
2.9 Gametogenesis
2.11 Oogenesis
2.13 The menstrual cycle
2.19 Pregnancy and the placenta

Testes within the scrotum, outside the abdominal cavity

Vas deferens looping over the urinary bladder

Seminal vesicle, prostate and bulbourethral glands

Urethra running through the penis to the urethral meatus

Figure 2.A Diagrammatic sectional view of male pelvis showing reproductive system

Testicular lobules, about 250 of them in each testis

Seminiferous tubules, one to three in each lobule

Rete testis and vasa efferentia leaving the testis

Epididymis along the posterior surface, leading to the vas deferens

Figure 2.B Diagrammatic view of male reproductive system (part of testis is open to show inner details)

2.1 The testis

Where it sits, and why

- the testes are situated outside the abdominal cavity within a pouch called the scrotum
- the scrotum helps maintain the low temperature of the testes, 2 to 2.5 degrees C lower than the normal internal body temperature, which is necessary for spermatogenesis
- in adults each testis is oval in shape, with a length of about 4 to 5 cm and a width of about 2 to 3 cm, and is covered by a dense covering
- each testis has about 250 compartments called testicular lobules

Inside a seminiferous tubule

- each lobule contains one to three highly coiled seminiferous tubules in which sperms are produced
- each seminiferous tubule is lined on its inside by two types of cell, the male germ cells or spermatogonia and the Sertoli cells, the germ cells undergoing meiotic divisions finally leading to sperm formation while the Sertoli cells provide nutrition to them
- the regions outside the seminiferous tubules, called interstitial spaces, contain small blood vessels and interstitial or Leydig cells, which synthesise and secrete testicular hormones called androgens
- other immunologically competent cells are also present

2.2 The male sex accessory ducts

From testis to urethra

- the male sex accessory ducts include the rete testis, vasa efferentia, epididymis and vas deferens
- the seminiferous tubules of the testis open into the vasa efferentia through the rete testis, and the vasa efferentia leave the testis and open into the epididymis located along the posterior surface of each testis
- the epididymis leads to the vas deferens, which ascends to the abdomen and loops over the urinary bladder, receives a duct from the seminal vesicle and opens into the urethra as the ejaculatory duct
- these ducts store and transport the sperms from the testis to the outside through the urethra, which originates from the urinary bladder and extends through the penis to its external opening, the urethral meatus

Spermatogonia, the male germ cells lining the tubule

Sertoli cells among them, feeding the germ cells

Interstitial spaces outside the tubule

Leydig cells there, secreting the androgens

Figure 2.C Diagrammatic sectional view of seminiferous tubule

2.3 Penis and accessory glands

The external genitalia

- the penis is the male external genitalia
- it is made up of special tissue that helps in erection of the penis to facilitate insemination
- the enlarged end of the penis, called the glans penis, is covered by a loose fold of skin called the foreskin

The accessory glands

- they include paired seminal vesicles, a prostate and paired bulbourethral glands
- secretions of these glands constitute the seminal plasma, which is rich in fructose, calcium and certain enzymes
- the secretions of the bulbourethral glands also help in the lubrication of the penis

2.4 The female reproductive system

The parts, and what they support

- the system consists of a pair of ovaries along with a pair of oviducts, uterus, cervix, vagina and the external genitalia, located in the pelvic region
- these parts, along with a pair of mammary glands, are integrated structurally and functionally to support the processes of ovulation, fertilisation, pregnancy, birth and child care

The ovaries

- the ovaries are the primary female sex organs, producing the female gamete, the ovum, and several steroid hormones, the ovarian hormones
- they are located one on each side of the lower abdomen, each about 2 to 4 cm in length and connected to the pelvic wall and uterus by ligaments
- each ovary is covered by a thin epithelium which encloses the ovarian stroma, and the stroma is divided into two zones, a peripheral cortex and an inner medulla

Ovaries on either side of the lower abdomen

Uterus in the midline, opening through the cervix

Vagina below it, forming the birth canal

External genitalia at the outlet

Figure 2.D Diagrammatic sectional view of female pelvis showing reproductive system

Infundibulum with its fimbriae, next to the ovary

Ampulla, the wider part of the oviduct

Isthmus, narrow, joining the uterus

Cervical canal, leading from uterus to vagina

Figure 2.E Diagrammatic sectional view of the female reproductive system

2.5 The female accessory ducts

The oviducts

- the oviducts or fallopian tubes, the uterus and the vagina constitute the female accessory ducts
- each fallopian tube is about 10-12 cm long and extends from the periphery of each ovary to the uterus
- the part closer to the ovary is the funnel-shaped infundibulum, whose edges possess finger-like projections called fimbriae, which help in collection of the ovum after ovulation
- the infundibulum leads to a wider part of the oviduct called the ampulla, and the last part, the isthmus, has a narrow lumen and joins the uterus

Mammary lobes, 15 to 20 in each breast

Alveoli in clusters, where milk is secreted and stored

Mammary tubules joining into a mammary duct

Mammary ampulla and lactiferous duct, opening at the nipple

Figure 2.F A diagrammatic sectional view of Mammary gland

2.6 The uterus and the external genitalia

The uterus

- the uterus is single and is also called the womb, its shape like an inverted pear, and it is supported by ligaments attached to the pelvic wall
- it opens into the vagina through a narrow cervix, and the cavity of the cervix is called the cervical canal, which along with the vagina forms the birth canal
- the wall of the uterus has three layers of tissue, the external thin membranous perimetrium, the middle thick layer of smooth muscle called the myometrium, and the inner glandular layer called the endometrium, which lines the uterine cavity
- the endometrium undergoes cyclical changes during the menstrual cycle, while the myometrium exhibits strong contraction during delivery of the baby

The external genitalia

- they include the mons pubis, labia majora, labia minora, hymen and clitoris
- the mons pubis is a cushion of fatty tissue covered by skin and pubic hair, the labia majora are fleshy folds of tissue extending down from it and surrounding the vaginal opening, and the labia minora are paired folds of tissue under the labia majora
- the opening of the vagina is often covered partially by a membrane called the hymen, and the clitoris is a tiny finger-like structure lying at the upper junction of the two labia minora above the urethral opening

About the hymen

- the hymen is often torn during the first coitus, that is intercourse
- it can also be broken by a sudden fall or jolt, by insertion of a vaginal tampon, or by active participation in some sports such as horseback riding and cycling
- in some women the hymen persists even after coitus, and in fact the presence or absence of the hymen is not a reliable indicator of virginity or sexual experience

Spermatogonium, diploid, with 46 chromosomes

Primary spermatocyte, entering the first meiotic division

Secondary spermatocytes, haploid, with 23 chromosomes each

Spermatids, four of them, transformed into spermatozoa

Figure 2.G Diagrammatic sectional view of a seminiferous tubule (enlarged)

2.7 The mammary glands

How a breast is built

- a functional mammary gland is characteristic of all female mammals, the mammary glands being paired structures, the breasts, that contain glandular tissue and a variable amount of fat
- the glandular tissue of each breast is divided into 15-20 mammary lobes containing clusters of cells called alveoli, whose cells secrete milk, stored in the cavities or lumens of the alveoli
- the alveoli open into mammary tubules, the tubules of each lobe join to form a mammary duct, and several mammary ducts join to form a wider mammary ampulla, connected to the lactiferous duct and opening at the nipple, from where milk is expressed during breastfeeding

2.8 Gametogenesis and spermatogenesis

From spermatogonium to spermatid

- the primary sex organs, the testis in males and the ovaries in females, produce the gametes, sperms and ovum respectively, by the process called gametogenesis
- in the testis the immature male germ cells or spermatogonia produce sperms by spermatogenesis, which begins at puberty, the spermatogonia present on the inside wall of the seminiferous tubules multiplying by mitotic division, each being diploid and containing 46 chromosomes
- some of the spermatogonia, called primary spermatocytes, periodically undergo meiosis, and a primary spermatocyte completes the first meiotic division, the reduction division, leading to two equal haploid cells called secondary spermatocytes, which have only 23 chromosomes each
- the secondary spermatocytes undergo the second meiotic division to produce four equal haploid spermatids, which leaves the number of chromosomes in a spermatid to be worked out

Spermiogenesis, and the hormones behind it

- the spermatids are transformed into spermatozoa, that is sperms, by the process called spermiogenesis, after which the sperm heads become embedded in the Sertoli cells and are finally released from the seminiferous tubules by the process called spermiation
- spermatogenesis starts at the age of puberty due to a significant increase in the secretion of gonadotropin releasing hormone (GnRH), a hypothalamic hormone
- the increased levels of GnRH act at the anterior pituitary gland and stimulate secretion of two gonadotropins, luteinising hormone (LH) and follicle stimulating hormone (FSH)
- LH acts at the Leydig cells and stimulates the synthesis and secretion of androgens, which in turn stimulate spermatogenesis

Head, with an elongated haploid nucleus

Acrosome, the cap over the front of the head

Neck, then the middle piece packed with mitochondria

Tail, whose movement gives the sperm its motility

Figure 2.H Structure of a sperm

2.9 The sperm

Its structure

- FSH acts on the Sertoli cells and stimulates secretion of some factors which help in spermiogenesis
- a sperm is a microscopic structure of head, neck, middle piece and tail, with a plasma membrane enveloping the whole body
- the head contains an elongated haploid nucleus, whose anterior portion is covered by a cap-like structure, the acrosome, filled with enzymes that help fertilisation of the ovum
- the middle piece possesses numerous mitochondria, which produce the energy for the movement of the tail that gives the motility essential for fertilisation

Semen

- the human male ejaculates about 200 to 300 million sperms during a coitus, of which, for normal fertility, at least 60 per cent must have normal shape and size and at least 40 per cent must show vigorous motility
- sperms released from the seminiferous tubules are carried by the accessory ducts, and secretions of the epididymis, vas deferens, seminal vesicle and prostate are essential for their maturation and motility
- the seminal plasma with the sperms constitutes the semen, and the functions of the male accessory ducts and glands are maintained by the testicular androgens

2.10 Oogenesis

It begins before birth

- the process of formation of a mature female gamete is called oogenesis, and it is markedly different from spermatogenesis
- it begins in the embryonic stage, when a couple of million gamete mother cells or oogonia form within each fetal ovary, and no more are formed or added after birth
- these cells start division, enter prophase-I of meiosis and are temporarily arrested there, and are then called primary oocytes
- each primary oocyte gets surrounded by a layer of granulosa cells and is called a primary follicle, many of which degenerate between birth and puberty, so that at puberty only 60,000-80,000 primary follicles are left in each ovary

Secondary and tertiary follicles

- the primary follicles get surrounded by more layers of granulosa cells and a new theca and are called secondary follicles
- a secondary follicle soon becomes a tertiary follicle, marked by a fluid filled cavity called the antrum, and the theca layer forms an inner theca interna and an outer theca externa
- it is at this stage that the primary oocyte within the tertiary follicle grows and completes its first meiotic division, an unequal division giving a large haploid secondary oocyte and a tiny first polar body

Primary follicle, then secondary and tertiary

Antrum, the fluid filled cavity of the tertiary follicle

Graafian follicle, mature and ready to rupture

Corpus luteum, what remains after ovulation

Figure 2.I Diagrammatic Section view of ovary

Spermatogonium to primary and secondary spermatocyte to four spermatids

Oogonium to primary oocyte, arrested in prophase-I

Secondary oocyte and first polar body, from an unequal division

Ootid and second polar body, after the second division

Figure 2.L Schematic representation of (a) Spermatogenesis; (b) Oogenesis

2.11 Ovulation

The Graafian follicle ruptures

- the secondary oocyte retains the bulk of the nutrient rich cytoplasm of the primary oocyte, which leaves the advantage of that arrangement to be worked out, and whether the first polar body divides further or degenerates is at present not certain
- the tertiary follicle further changes into the mature or Graafian follicle
- the secondary oocyte forms a new membrane called the zona pellucida surrounding it
- the Graafian follicle now ruptures to release the secondary oocyte, that is the ovum, from the ovary, by the process called ovulation

2.12 The menstrual cycle

Menarche onwards

- the reproductive cycle in the female primates, such as monkeys, apes and human beings, is called the menstrual cycle
- the first menstruation begins at puberty and is called menarche
- in human females menstruation is repeated at an average interval of about 28/29 days, and the cycle of events starting from one menstruation till the next is the menstrual cycle
- one ovum is released, that is ovulation, during the middle of each menstrual cycle

Menstrual phase, the flow lasting 3 to 5 days

Follicular phase, the follicle growing and the endometrium regenerating

Ovulation at mid-cycle, driven by the LH surge

Luteal phase, the corpus luteum secreting progesterone

Figure 2.M Diagrammatic presentation of various events during a menstrual cycle

2.13 Phases of the cycle

The menstrual phase

- the cycle starts with the menstrual phase, when menstrual flow occurs, and it lasts for 3-5 days
- the menstrual flow results from breakdown of the endometrial lining of the uterus and its blood vessels, which forms a liquid that comes out through the vagina
- menstruation only occurs if the released ovum is not fertilised, so lack of menstruation may be indicative of pregnancy, although it may also be caused by other underlying causes such as stress and poor health

The follicular phase

- the menstrual phase is followed by the follicular phase
- during this phase the primary follicles in the ovary grow to become a fully mature Graafian follicle, and at the same time the endometrium of the uterus regenerates through proliferation
- these changes in the ovary and the uterus are induced by changes in the levels of pituitary and ovarian hormones

An ovum, surrounded by a few sperms

Zona pellucida, the layer they must cross

Acrosomal secretions opening the way for one sperm

Figure 2.N Ovum surrounded by few sperms

2.14 Ovulation and the luteal phase

The LH surge and after

- the secretion of gonadotropins, LH and FSH, increases gradually during the follicular phase and stimulates follicular development and the secretion of estrogens by the growing follicles, both attaining a peak in the middle of the cycle, about the 14th day
- rapid secretion of LH leading to its maximum level during the mid-cycle, called the LH surge, induces rupture of the Graafian follicle and thereby the release of the ovum, that is ovulation
- the ovulatory phase is followed by the luteal phase, in which the remaining parts of the Graafian follicle transform into the corpus luteum, secreting large amounts of progesterone, essential for maintaining the endometrium and so for implantation of the fertilised ovum and the other events of pregnancy
- during pregnancy all events of the cycle stop and there is no menstruation, while without fertilisation the corpus luteum degenerates, the endometrium disintegrates and menstruation marks a new cycle, and menstrual cycles cease around 50 years of age, which is termed menopause, cyclic menstruation being an indicator of the normal reproductive phase between menarche and menopause

2.15 Fertilisation and implantation

Where fertilisation happens

- during copulation, that is coitus, semen is released by the penis into the vagina, which is insemination
- the motile sperms swim rapidly, pass through the cervix, enter the uterus and reach the ampullary region of the fallopian tube, where the ovum released by the ovary is also carried, and there fertilisation takes place
- fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region, which is why not all copulations lead to fertilisation and pregnancy
- the fusion of a sperm with an ovum is fertilisation, and during it a sperm comes in contact with the zona pellucida layer of the ovum and induces changes in the membrane that block the entry of further sperms, so only one sperm can fertilise an ovum

Menstrual hygiene

- maintenance of hygiene and sanitation during menstruation is very important, so take bath and clean yourself regulary
- use sanitary napkins or clean homemade pads, and change them after every 4 to 5 hrs as per the requirement
- dispose of the used sanitary napkins properly, wrapping them with a used paper, and do not throw them in the drainpipe of toilets or in the open area
- after handling the napkin wash hands with soap

Ovum released from the ovary, taken up by the fimbriae

Fertilisation in the ampullary region of the tube

Cleavage, the zygote dividing as it moves along

The growing embryo passing towards the uterus

Figure 2.0 Transport of ovum, fertilisation and passage of growing embryo through fallopian tube

2.16 Sex determination and cleavage

The zygote forms

- the secretions of the acrosome help the sperm enter the cytoplasm of the ovum through the zona pellucida and the plasma membrane, and this induces the completion of the meiotic division of the secondary oocyte
- the second meiotic division is also unequal and results in a second polar body and a haploid ovum, the ootid
- soon the haploid nucleus of the sperm and that of the ovum fuse together to form a diploid zygote, which leaves the number of chromosomes in the zygote to be worked out

How sex is decided

- the sex of the baby has been decided at this stage itself
- the chromosome pattern in the human female is XX and that in the male is XY, so all the haploid gametes produced by the female have the sex chromosome X, whereas in the male gametes the sex chromosome could be either X or Y, 50 per cent of sperms carrying the X chromosome and the other 50 per cent the Y
- after fusion of the male and female gametes the zygote carries either XX or XY, depending on whether the sperm carrying X or Y fertilised the ovum, and the zygote carrying XX develops into a female baby and XY into a male
- that is why it is scientifically correct to say that the sex of the baby is determined by the father and not by the mother

Cleavage

- the mitotic division starts as the zygote moves through the isthmus of the oviduct towards the uterus, and this division is called cleavage
- it forms 2, 4, 8 and 16 daughter cells called blastomeres

Foetus within the uterus, surrounded by uterine tissue

Chorionic villi interdigitating with that tissue

Placenta, the unit they form together

Umbilical cord, connecting placenta to embryo

Figure 2.P The human foetus within the uterus

2.17 Implantation

Morula to blastocyst

- the embryo with 8 to 16 blastomeres is called a morula
- the morula continues to divide and transforms into a blastocyst as it moves further into the uterus
- the blastomeres in the blastocyst are arranged into an outer layer called the trophoblast and an inner group of cells attached to it called the inner cell mass
- the trophoblast layer then gets attached to the endometrium and the inner cell mass gets differentiated as the embryo, after which the uterine cells divide rapidly and cover the blastocyst, so that it becomes embedded in the endometrium, which is implantation, and it leads to pregnancy

2.18 Pregnancy and the placenta

How the placenta forms and what it does

- after implantation, finger-like projections appear on the trophoblast, called chorionic villi, which are surrounded by the uterine tissue and maternal blood
- the chorionic villi and uterine tissue become interdigitated with each other and jointly form a structural and functional unit between the developing embryo or foetus and the maternal body, called the placenta
- the placenta facilitates the supply of oxygen and nutrients to the embryo and also the removal of carbon dioxide and excretory/waste materials produced by it, and it is connected to the embryo through an umbilical cord which helps in the transport of substances to and from the embryo

The hormones of pregnancy

- the placenta also acts as an endocrine tissue and produces several hormones such as human chorionic gonadotropin (hCG), human placental lactogen (hPL), estrogens and progestogens
- in the later phase of pregnancy a hormone called relaxin is also secreted by the ovary, and hCG, hPL and relaxin are produced in women only during pregnancy
- during pregnancy the levels of other hormones such as estrogens, progestogens, cortisol, prolactin and thyroxine are increased several-fold in the maternal blood
- increased production of these hormones is essential for supporting fetal growth, the metabolic changes in the mother and the maintenance of pregnancy

2.19 Embryonic development

The three germ layers

- immediately after implantation the inner cell mass, that is the embryo, differentiates into an outer layer called ectoderm and an inner layer called endoderm
- a mesoderm soon appears between the ectoderm and the endoderm, and these three layers give rise to all the tissues and organs in adults
- the inner cell mass contains certain cells called stem cells, which have the potency to give rise to all the tissues and organs

Month by month

- human pregnancy lasts 9 months, which leaves the length of pregnancy in dogs, elephants and cats to be found out
- after one month of pregnancy the embryo's heart is formed, and the first sign of a growing foetus may be noticed by listening to the heart sound carefully through the stethoscope, while by the end of the second month the foetus develops limbs and digits
- by the end of 12 weeks, that is the first trimester, most of the major organ systems are formed, the limbs and external genital organs being well-developed, and the first movements of the foetus and appearance of hair on the head are usually observed during the fifth month
- by the end of about 24 weeks, that is the end of the second trimester, the body is covered with fine hair, the eye-lids separate and eyelashes are formed, and by the end of nine months the foetus is fully developed and ready for delivery

2.20 Parturition and lactation

Parturition

- the average duration of human pregnancy is about 9 months, which is called the gestation period, and vigorous contraction of the uterus at the end of pregnancy causes expulsion or delivery of the foetus, a process called parturition
- parturition is induced by a complex neuroendocrine mechanism, the signals originating from the fully developed foetus and the placenta, which induce mild uterine contractions called the foetal ejection reflex
- this triggers release of oxytocin from the maternal pituitary, which acts on the uterine muscle and causes stronger uterine contractions, which in turn stimulate further secretion of oxytocin, and this stimulatory reflex continues until the baby is expelled out of the uterus through the birth canal
- soon after the infant is delivered the placenta is also expelled out of the uterus, which leaves what doctors inject to induce delivery to be worked out

Lactation

- the mammary glands of the female undergo differentiation during pregnancy and start producing milk towards the end of pregnancy, by the process called lactation
- this helps the mother in feeding the new-born
- the milk produced during the initial few days of lactation is called colostrum, which contains several antibodies absolutely essential to develop resistance in new-born babies
- breast-feeding during the initial period of infant growth is recommended by doctors for bringing up a healthy baby

2.21 Summary

The male system

- humans are sexually reproducing and viviparous, and the male reproductive system is composed of a pair of testes, the male sex accessory ducts, the accessory glands and the external genitalia
- each testis has about 250 compartments called testicular lobules, and each lobule contains one to three highly coiled seminiferous tubules, each lined inside by spermatogonia and Sertoli cells
- the spermatogonia undergo meiotic divisions leading to sperm formation, while Sertoli cells provide nutrition to the dividing germ cells, and the Leydig cells outside the seminiferous tubules synthesise and secrete testicular hormones called androgens, the male external genitalia being the penis

The female system

- it consists of a pair of ovaries, a pair of oviducts, a uterus, a vagina, external genitalia and a pair of mammary glands
- the ovaries produce the female gamete, the ovum, and some steroid hormones, the ovarian hormones, and ovarian follicles in different stages of development are embedded in the stroma
- the oviducts, uterus and vagina are the female accessory ducts, the uterus having three layers, the perimetrium, myometrium and endometrium
- the female external genitalia includes the mons pubis, labia majora, labia minora, hymen and clitoris, and the mammary glands are one of the female secondary sexual characteristics

Gametes, cycle and birth

- spermatogenesis results in the formation of sperms that are transported by the male sex accessory ducts, a normal human sperm being composed of a head, neck, middle piece and tail, while the process of formation of mature female gametes is called oogenesis
- the reproductive cycle of female primates is called the menstrual cycle, which starts only after attaining sexual maturation, that is puberty, one ovum being released per cycle during ovulation, and the cyclical changes in the ovary and the uterus are induced by changes in the levels of pituitary and ovarian hormones
- after coitus sperms are transported to the ampulla, where the sperm fertilises the ovum leading to a diploid zygote, the presence of X or Y chromosome in the sperm determining the sex of the embryo, and the zygote undergoes repeated mitotic division to form a blastocyst, which is implanted in the uterus resulting in pregnancy
- after nine months the fully developed foetus is ready for delivery, and childbirth, called parturition, is induced by a complex neuroendocrine mechanism involving cortisol, estrogens and oxytocin, while mammary glands differentiate during pregnancy and secrete milk after child-birth, the new-born being fed by lactation during the initial few months of growth

2.22 Exercises

From this chapter

1 Fill in the blanks: (a) Humans reproduce _____ (asexually/sexually) (b) Humans are _____ (oviparous, viviparous, ovoviviparous) (c) Fertilisation is _____ in humans (external/internal) (d) Male and female gametes are _____ (diploid/haploid) (e) Zygote is _____ (diploid/haploid)

2.23 Exercises, continued

From this chapter, continued

1 (f) The process of release of ovum from a mature follicle is called _____ (g) Ovulation is induced by a hormone called _____ (h) The fusion of male and female gametes is called _____ (i) Fertilisation takes place in _____ (j) Zygote divides to form _____which is implanted in uterus. (k) The structure which provides vascular connection between foetus and uterus is called _____
2 Draw a labelled diagram of male reproductive system.
3 Draw a labelled diagram of female reproductive system.
4 Write two major functions each of testis and ovary.
5 Describe the structure of a seminiferous tubule.
6 What is spermatogenesis? Briefly describe the process of spermatogenesis.
7 Name the hormones involved in regulation of spermatogenesis.
8 Define spermiogenesis and spermiation.
9 Draw a labelled diagram of sperm.
10 What are the major components of seminal plasma?
11 What are the major functions of male accessory ducts and glands?
12 What is oogenesis? Give a brief account of oogenesis.
13 Draw a labelled diagram of a section through ovary.
14 Draw a labelled diagram of a Graafian follicle?
15 Name the functions of the following: (a) Corpus luteum (b) Endometrium (c) Acrosome (d) Sperm tail (e) Fimbriae
16 Identify True/False statements. Correct each false statement to make it true. (a) Androgens are produced by Sertoli cells. (True/False) (b) Spermatozoa get nutrition from Sertoli cells. (True/False) (c) Leydig cells are found in ovary. (True/False) (d) Leydig cells synthesise androgens. (True/False) (e) Oogenesis takes place in corpus luteum. (True/False) (f) Menstrual cycle ceases during pregnancy. (True/False) (g) Presence or absence of hymen is not a reliable indicator of virginity or sexual experience. (True/False)
17 What is menstrual cycle? Which hormones regulate menstrual cycle?
18 What is parturition? Which hormones are involved in induction of parturition?
19 In our society the women are often blamed for giving birth to daughters. Can you explain why this is not correct?
20 How many eggs are released by a human ovary in a month? How many eggs do you think would have been released if the mother gave birth to identical twins? Would your answer change if the twins born were fraternal?
21 How many eggs do you think were released by the ovary of a female dog which gave birth to 6 puppies?